POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2003A | Semester I / II | Practicum | 4.5 Credits

# Chemistry for Engineering Practices

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

| | |
|---|---|
| Programmes | **All-programme chemistry foundation group listed in the official syllabus** |
| Contact hours | **90** |
| Teaching scheme | **3:0:3:0** |
| Credits | **4.5** |
| CIE / ESE | **Theory 20 + Practical 20 / Theory 60** |
| Self-learning | **6 h/week; 90 h total** |

CURRICULUM INTEGRITY

## Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2003A**, titled **Chemistry for Engineering Practices**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

**Source treatment:** PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

**Verified source note:** No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

## What You Are Expected to Learn

**90**
Contact hours

**4.5**
Credits

**Theory 20 + Practical 20**
CIE

**Theory 60**
ESE

### Course introduction

Chemistry for Engineering Practices integrates basic chemical concepts with laboratory habits for engineering materials, water, electrochemistry, corrosion and sustainability. It develops accuracy in solution preparation and analysis as well as informed material selection.

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### Prerequisite foundation

Basic knowledge of Chemistry (Secondary Education).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

**Teaching and assessment scheme**

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE | ESE |
|---|---|---|---|---|---|---|---|---|
| 3 | 0 | 3 | 0 | 6 h/week; 90 h total | 90 | 4.5 | Theory 20 + Practical 20 | Theory 60 |

STUDY METHOD

## How to Use This Handbook

### 1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

### 4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

## Objectives and Course Outcomes

### Official course objectives

1. Relate atoms, solutions and thermodynamics to engineering applications.
2. Apply analytical chemistry, water treatment, acids and bases.
3. Develop electrochemistry and corrosion understanding for engineering systems.
4. Select materials and explain green chemistry principles.

### Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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**Official Course Outcomes**

| CO | Official outcome | Bloom level |
|---|---|---|
| CO1 | Apply basic concepts of atoms, molecules, solutions and thermodynamic processes in scientific and engineering contexts. | Apply |
| CO2 | Apply analytical chemistry, water treatment and acid-base concepts to engineering problems. | Apply |
| CO3 | Understand electrochemistry and corrosion and apply it to engineering systems. | Apply |
| CO4 | Explore engineering materials and explain green-chemistry principles for sustainable practice. | Apply |

| Official CO-PO articulation matrix | | | | | | | |
|---|---|---|---|---|---|---|---|
| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7–PO11 |
| CO1 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO2 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO3 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO4 | 3 | 2 | 2 | 2 | 2 | – | – |

SYLLABUS COVERAGE

## Curriculum Coverage Map

| All official major topics mapped to handbook chapters | | | | | |
|---|---|---|---|---|---|
| Module | Title | Official major topics | Hours | CO | Handbook section |
| 1 | Fundamentals, Solutions and Thermodynamics | Atoms and molecules; solution concentration; systems, processes and thermodynamic functions; preparation of standard solutions. | 11 L + 9 P hours | CO1 | Module 1 |
| 2 | Analytical Chemistry, Water and Acids/Bases | Volumetric analysis; hardness and water treatment; potable water; pH and pOH; standardisation and estimation. | 12 L + 12 P hours | CO2 | Module 2 |
| 3 | Electrochemistry, Corrosion and Batteries | Redox reactions; conductors; electrolysis; Faraday laws; electrochemical cells; corrosion prevention; emerging batteries. | 11 L + 9 P hours | CO3 | Module 3 |
| 4 | Engineering Materials and Green Chemistry | Alloys; polymers and rubbers; nanomaterials; MOFs; biomaterials; biofuel; green chemistry. | 11 L + 9 P hours | CO4 | Module 4 |

LEARNING ROADMAP

## How the Modules Connect

### Fundamentals, Solutions and Thermodynamics

Atoms and molecules; solution concentration; systems, processes and thermodynamic functions; preparation of standard solutions.

CO1 · 11 L + 9 P hours

### Analytical Chemistry, Water and Acids/Bases

Volumetric analysis; hardness and water treatment; potable water; pH and pOH; standardisation and estimation.

CO2 · 12 L + 12 P hours

### Electrochemistry, Corrosion and Batteries

Redox reactions; conductors; electrolysis; Faraday laws; electrochemical cells; corrosion prevention; emerging batteries.

CO3 · 11 L + 9 P hours

### Engineering Materials and Green Chemistry

Alloys; polymers and rubbers; nanomaterials; MOFs; biomaterials; biofuel; green chemistry.

CO4 · 11 L + 9 P hours

**MODULE 1**

## Fundamentals, Solutions and Thermodynamics

Atoms and molecules; solution concentration; systems, processes and thermodynamic functions; preparation of standard solutions.

**11 L + 9 P hours** **CO1** **Understand · Apply**

### Learning goals

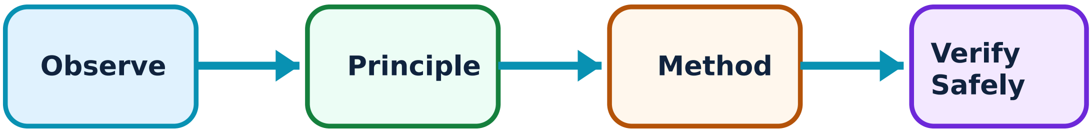
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Fundamentals, Solutions and Thermodynamics: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Atoms, particles and atomic identifiers−

An atom is the smallest chemical unit retaining the identity of an element; a molecule is an associated group of atoms. Electrons carry negative charge, protons positive charge and neutrons no charge. Atomic number identifies protons, while mass number represents protons plus neutrons.

### Study and application method

For an atom up to Z = 20, use atomic number to identify protons and electrons in a neutral atom, and subtract atomic number from mass number to obtain neutrons.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Mass number is not the same as average atomic mass shown in a periodic table.

## 2. Solutions, molarity, normality and ppm−

A solution contains solute dispersed in solvent. A standard solution has accurately known concentration. Molarity expresses moles per litre; normality expresses gram-equivalents per litre; ppm is useful for very dilute quantities. The selected unit must match the analytical purpose.

### Study and application method

Read target concentration and final volume first, calculate required amount, dissolve safely and make up to mark only after complete transfer.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

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**Common mistake / precaution:** A volumetric flask gives a final accurate volume; it must not be used as a heating vessel.

## 3. Thermodynamic systems and processes−

A system is the selected part for study; surroundings are everything else relevant to it. Open, closed and isolated systems differ by matter/energy exchange. Isothermal, isobaric, isochoric and adiabatic describe constraints. Internal energy, enthalpy and entropy are state functions used to describe change.

### Study and application method

Use a boundary sketch and name controlled variable. Differentiate reversible idealisation from an irreversible real process, then classify spontaneous, endothermic or exothermic behaviour carefully.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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**Common mistake / precaution:** Spontaneous does not mean instant; it refers to thermodynamic tendency under stated conditions.

## 4. Solution-preparation and thermal-reaction practice−

The official work includes 0.1 N oxalic acid, 0.1 N sodium carbonate, 0.1 M zinc sulphate and 0.1 M magnesium sulphate, along with endothermic/exothermic demonstrations. Each record requires calculation, weighing, dissolution, transfer, make-up and label.

### Study and application method

Wear PPE, read labels, use clean glassware and document observations such as temperature change rather than merely writing that reaction occurred.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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**Common mistake / precaution:** Never taste chemicals or return excess reagent to a stock bottle.

### Molarity calculator

**Moles of solute**

**0.1**

**Solution volume (L)**

**1**

**Enter valid values and choose Calculate.**

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

## Analytical Chemistry, Water and Acids/Bases

Volumetric analysis; hardness and water treatment; potable water; pH and pOH; standardisation and estimation.

12 L + 12 P hours | CO2 | Understand · Apply

### Learning goals

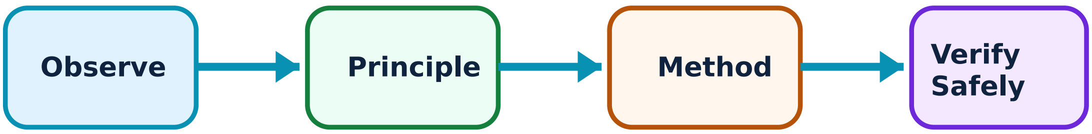
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Analytical Chemistry, Water and Acids/Bases: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Volumetric analysis and indicators−

Titration finds concentration by reacting measured volumes to an endpoint identified using a suitable indicator. Standard solution, endpoint and indicator choice are different ideas. The normality relation $V_1N_1 = V_2N_2$ expresses equivalent reaction at completion for the treatments prescribed.

### Study and application method

Write balanced reaction where required, rinse apparatus with correct solution, take concordant readings and record initial/final burette values.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Observe colour change at endpoint, not after substantial excess titrant is added.

## 2. Hardness, potable water and treatment−

Hard water contains dissolved salts that interfere with soap action; temporary and permanent hardness have different origin/treatment. Potable water needs acceptable physical, chemical and biological characteristics. Municipal treatment proceeds through screening, sedimentation, coagulation, filtration and sterilisation; RO and nanofiltration are filtration methods in the syllabus.

### Study and application method

Use flow arrows in correct process order. Separate suspended-solid removal from disinfection of microorganisms.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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**Common mistake / precaution:** Filtration alone does not automatically make all water potable; treatment context matters.

## 3. Arrhenius acids, bases, pH and pOH−

Arrhenius acids produce hydrogen ions in aqueous solution and bases produce hydroxide ions. pH and pOH are logarithmic measures used to express acidic or basic character. Their relation is applied for simple aqueous problems in the official scope.

### Study and application method

State whether a quantity is pH, pOH or hydrogen-ion-related before calculation. Use pH scale as interpretation tool and report only justified precision.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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**Common mistake / precaution:** pH is not a direct linear concentration scale; a change of one pH unit has logarithmic meaning.

## 4. Quantitative and qualitative analysis practice−

Official activities include HCl and EDTA standardisation, estimation of NaOH/KOH, total hardness by EDTA and pH determination by meter, universal indicator and paper. Practical skill means correct apparatus use, clean endpoints and defensible data.

### Study and application method

Write aim, apparatus, reaction/principle, procedure, observations, calculation, result and precautions in record.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Acids and bases can injure skin/eyes; follow laboratory instruction for addition and spill response.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

## Electrochemistry, Corrosion and Batteries

Redox reactions; conductors; electrolysis; Faraday laws; electrochemical cells; corrosion prevention; emerging batteries.

11 L + 9 P hours | CO3 | Understand · Apply

### Learning goals

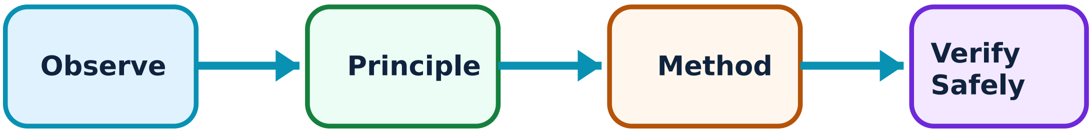
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Electrochemistry, Corrosion and Batteries: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Redox, conductors and electrolysis−

Oxidation and reduction occur together in a redox reaction. Metallic conductors carry current through electrons; electrolytic conductors use ions. An electrolytic cell uses external energy to drive chemical change. Faraday’s laws relate amount of chemical change to charge passed.

#### Study and application method

Identify oxidation and reduction separately before discussing the complete reaction. In calculations track current, time and charge with correct units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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**Common mistake / precaution:** An electrolyte is not merely a liquid; it must provide mobile ions under stated conditions.

## 2. Electrochemical cells and applications−

A Daniel cell illustrates conversion of chemical energy to electrical energy and uses a salt bridge to maintain charge balance. Primary, secondary and fuel cells are classified by their mode of use. Electroplating and electrolytic refining are official applications.

### Study and application method

Draw both half-cells, electrode labels, external electron path and salt-bridge role. Explain an application as controlled process sequence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

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**Malayalam support:** □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**Common mistake / precaution:** Salt bridge maintains ionic balance; it is not intended as the main electron path.

## 3. Corrosion and prevention−

Corrosion is deterioration of metal through chemical or electrochemical interaction with environment. Moisture, oxygen, temperature, impurities and galvanic contact can affect rate. Barrier coatings, metallic/non-metallic protection and sacrificial-anode protection are official approaches.

### Study and application method

Identify environment, substrate and consequence of coating damage. Explain why a sacrificial anode is selected to corrode preferentially.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** A coating is not automatically permanent protection; inspection and maintenance influence service life.

## 4. Electrochemistry laboratory practice−

The official work includes salt bridge and Daniel-cell construction, nickel coating of steel, conductivity observation and electrolysis of water. Their value is connecting observed current, electrode process and chemical change.

### Study and application method

Inspect leads, use appropriate DC supply, label anode/cathode according to cell type and ventilate as directed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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**Common mistake / precaution:** Electrolysis may generate gases; keep ignition sources away and follow instructor guidance.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

## Engineering Materials and Green Chemistry

Alloys; polymers and rubbers; nanomaterials; MOFs; biomaterials; biofuel; green chemistry.

**11 L + 9 P hours** **CO4** **Understand · Apply**

### Learning goals

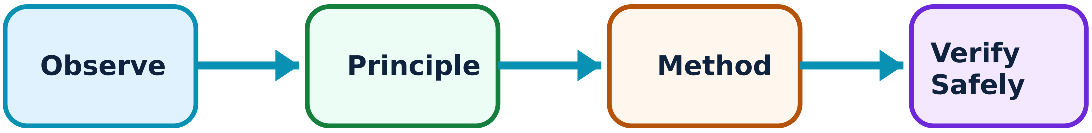
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Engineering Materials and Green Chemistry: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Alloys, polymers and rubber−

Alloying changes useful engineering properties. Polymer science distinguishes monomer, polymer, polymerisation, homopolymer and copolymer, then thermoplastics and thermosets. The official examples include PE, PVC, Nylon-66, Bakelite, PET, PU, PBT, PC and PMMA. Vulcanisation changes practical properties of natural rubber.

### Study and application method

Choose material by service requirement: strength, processing route, temperature, chemical exposure and end-of-life concern. Learn official examples with one application.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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**Common mistake / precaution:** Do not label every plastic recyclable or every polymer thermoplastic; structure and processing behaviour matter.

## 2. Nanomaterials, MOFs and biomaterials−

Nanomaterials have nanoscale features that can alter surface area and behaviour. Carbon nanotubes, fullerenes and graphene are examples. Metal–organic frameworks are porous advanced materials. Biomaterials can be metallic, ceramic, polymeric or composite.

### Study and application method

Connect structure to application carefully; high surface area explains some uses but is not a universal answer. Label material class in comparisons.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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**Common mistake / precaution:** Nano describes scale, not a guarantee that a material is safe or suitable for every use.

### 3. Biofuel and green chemistry−

Biofuels are fuels derived from biological sources. Green chemistry seeks prevention of waste and safer, resource-conscious chemical design. The syllabus asks for principles, importance and applications.

## Study and application method

For a process, identify feedstock, energy demand, waste, hazards and potential improvement. Use green-chemistry principles as a checklist.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

## Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
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**Common mistake / precaution:** Sustainable lab practice includes segregation and hazard control; using less of a hazardous reagent is not automatically sufficient.

## 4. Materials laboratory practice−

Official experiments include estimation of copper/zinc in brass, iron in ore and preparation of urea-formaldehyde and phenol-formaldehyde resins. A record connects chemical principle, observations and calculated result.

### Study and application method

Plan sequence, label samples, keep raw-data table and discuss uncertainty before finalising result.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Resin preparation and metal analysis require PPE, ventilation and supervised waste disposal.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

## Rapid Recall Bank

### Recall 1

State symbols and SI units before substitution.

### Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result

→ precautions.

### Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

## Diagram Library for Rapid Revision

### Module 1: Fundamentals, Solutions and Thermodynamics

Use the concept flow to revise observation, principle, method and verification.

### Module 2: Analytical Chemistry, Water and Acids/Bases

Use the concept flow to revise observation, principle, method and verification.

### Module 3: Electrochemistry, Corrosion and Batteries

Use the concept flow to revise observation, principle, method and verification.

### Module 4: Engineering Materials and Green Chemistry

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Prepare atom, solution and thermodynamics assignments and record-work analysis.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Practise indicator selection, hardness/pH observation and municipal-treatment flowcharts.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Prepare reports on redox reactions, Daniel cell, batteries and corrosion-related topics.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Develop materials, biomaterials, MOF, biofuel and green-synthesis study reports.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

# Assessment Methodology and Examination Pattern

| Official assessment structure | | |
|---|---|---|
| **Component** | **Official mark / conversion** | **Student evidence** |
| Attendance | 5 marks | Regular attendance and punctuality. |
| Continuous evaluation | Theory 20 + Practical 20 | Preparation, setup, observation, analysis, viva, record and discipline. |
| Practical tests | As stated in supplied PDF | Procedure/write-up, execution, result, viva and certified record. |
| Open-ended experiment | Where listed in supplied PDF | Originality, plan, execution/data, analysis/presentation and teamwork. |
| End-semester assessment | Theory 60 | Safe completion, record and viva according to official scheme. |

### Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

### Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

# Module-wise Questions with Answers

## Module 1 — Fundamentals, Solutions and Thermodynamics

**Explain Atoms, particles and atomic identifiers and state one correct method, application or precaution.–**

An atom is the smallest chemical unit retaining the identity of an element; a molecule is an associated group of atoms. Electrons carry negative charge, protons positive charge and neutrons no charge. Atomic number identifies protons, while mass number represents protons plus neutrons.

**Answer framework:** For an atom up to Z = 20, use atomic number to identify protons and electrons in a neutral atom, and subtract atomic number from mass number to obtain neutrons.

**Important point:** Mass number is not the same as average atomic mass shown in a periodic table.

**Explain Solutions, molarity, normality and ppm and state one correct method, application or precaution.−**

A solution contains solute dispersed in solvent. A standard solution has accurately known concentration. Molarity expresses moles per litre; normality expresses gram-equivalents per litre; ppm is useful for very dilute quantities. The selected unit must match the analytical purpose.

**Answer framework:** Read target concentration and final volume first, calculate required amount, dissolve safely and make up to mark only after complete transfer.

**Important point:** A volumetric flask gives a final accurate volume; it must not be used as a heating vessel.

**Explain Thermodynamic systems and processes and state one correct method, application or precaution.−**

A system is the selected part for study; surroundings are everything else relevant to it. Open, closed and isolated systems differ by matter/energy exchange. Isothermal, isobaric, isochoric and adiabatic describe constraints. Internal energy, enthalpy and entropy are state functions used to describe change.

**Answer framework:** Use a boundary sketch and name controlled variable. Differentiate reversible idealisation from an irreversible real process, then classify spontaneous, endothermic or exothermic behaviour carefully.

**Important point:** Spontaneous does not mean instant; it refers to thermodynamic tendency under stated conditions.

**Explain Solution-preparation and thermal-reaction practice and state one correct method, application or precaution.−**

The official work includes 0.1 N oxalic acid, 0.1 N sodium carbonate, 0.1 M zinc sulphate and 0.1 M magnesium sulphate, along with endothermic/exothermic demonstrations. Each record requires calculation, weighing, dissolution, transfer, make-up and label.

**Answer framework:** Wear PPE, read labels, use clean glassware and document observations such as temperature change rather than merely writing that reaction occurred.

**Important point:** Never taste chemicals or return excess reagent to a stock bottle.

### Module 2 — Analytical Chemistry, Water and Acids/Bases

**Explain Volumetric analysis and indicators and state one correct method, application or precaution.−**

Titration finds concentration by reacting measured volumes to an endpoint identified using a suitable indicator. Standard solution, endpoint and indicator choice are different ideas. The normality relation $V_1N_1 = V_2N_2$ expresses equivalent reaction at completion for the treatments prescribed.

**Answer framework:** Write balanced reaction where required, rinse apparatus with correct solution, take concordant readings and record initial/final burette values.

**Important point:** Observe colour change at endpoint, not after substantial excess titrant is added.

**Explain Hardness, potable water and treatment and state one correct method, application or precaution.−**

Hard water contains dissolved salts that interfere with soap action; temporary and permanent hardness have different origin/treatment. Potable water needs acceptable physical, chemical and biological characteristics. Municipal treatment proceeds through screening, sedimentation, coagulation, filtration and sterilisation; RO and nanofiltration are filtration methods in the syllabus.

**Answer framework:** Use flow arrows in correct process order. Separate suspended-solid removal from disinfection of microorganisms.

**Important point:** Filtration alone does not automatically make all water potable; treatment context matters.

**Explain Arrhenius acids, bases, pH and pOH and state one correct method, application or precaution.−**

Arrhenius acids produce hydrogen ions in aqueous solution and bases produce hydroxide ions. pH and pOH are logarithmic measures used to express acidic or basic character. Their relation is applied for simple aqueous problems in the official scope.

**Answer framework:** State whether a quantity is pH, pOH or hydrogen-ion-related before calculation. Use pH scale as interpretation tool and report only justified precision.

**Important point:** pH is not a direct linear concentration scale; a change of one pH unit has logarithmic meaning.

**Explain Quantitative and qualitative analysis practice and state one correct method, application or precaution.−**

Official activities include HCl and EDTA standardisation, estimation of NaOH/KOH, total hardness by EDTA and pH determination by meter, universal indicator and paper. Practical skill means correct apparatus use, clean endpoints and defensible data.

**Answer framework:** Write aim, apparatus, reaction/principle, procedure, observations, calculation, result and precautions in record.

**Important point:** Acids and bases can injure skin/eyes; follow laboratory instruction for addition and spill response.

## Module 3 — Electrochemistry, Corrosion and Batteries

**Explain Redox, conductors and electrolysis and state one correct method, application or precaution.−**

Oxidation and reduction occur together in a redox reaction. Metallic conductors carry current through electrons; electrolytic conductors use ions. An electrolytic cell uses external energy to drive chemical change. Faraday’s laws relate amount of chemical change to charge passed.

**Answer framework:** Identify oxidation and reduction separately before discussing the complete reaction. In calculations track current, time and charge with correct units.

**Important point:** An electrolyte is not merely a liquid; it must provide mobile ions under stated conditions.

**Explain Electrochemical cells and applications and state one correct method, application or precaution.−**

A Daniel cell illustrates conversion of chemical energy to electrical energy and uses a salt bridge to maintain charge balance. Primary, secondary and fuel cells are classified by their mode of use. Electroplating and electrolytic refining are official applications.

**Answer framework:** Draw both half-cells, electrode labels, external electron path and salt-bridge role. Explain an application as controlled process sequence.

**Important point:** Salt bridge maintains ionic balance; it is not intended as the main electron path.

**Explain Corrosion and prevention and state one correct method, application or precaution.−**

Corrosion is deterioration of metal through chemical or electrochemical interaction with environment. Moisture, oxygen, temperature, impurities and galvanic contact can affect rate. Barrier coatings, metallic/non-metallic protection and sacrificial-anode protection are official approaches.

**Answer framework:** Identify environment, substrate and consequence of coating damage. Explain why a sacrificial anode is selected to corrode preferentially.

**Important point:** A coating is not automatically permanent protection; inspection and maintenance influence service life.

**Explain Electrochemistry laboratory practice and state one correct method, application or precaution.−**

The official work includes salt bridge and Daniel-cell construction, nickel coating of steel, conductivity observation and electrolysis of water. Their value is connecting observed current, electrode process and chemical change.

**Answer framework:** Inspect leads, use appropriate DC supply, label anode/cathode according to cell type and ventilate as directed.

**Important point:** Electrolysis may generate gases; keep ignition sources away and follow instructor guidance.

## Module 4 — Engineering Materials and Green Chemistry

**Explain Alloys, polymers and rubber and state one correct method, application or precaution.−**

Alloying changes useful engineering properties. Polymer science distinguishes monomer, polymer, polymerisation, homopolymer and copolymer, then thermoplastics and thermosets. The official examples include PE, PVC, Nylon-66, Bakelite, PET, PU, PBT, PC and PMMA. Vulcanisation changes practical properties of natural rubber.

**Answer framework:** Choose material by service requirement: strength, processing route, temperature, chemical exposure and end-of-life concern. Learn official examples with one application.

**Important point:** Do not label every plastic recyclable or every polymer thermoplastic; structure and processing behaviour matter.

**Explain Nanomaterials, MOFs and biomaterials and state one correct method, application or precaution.−**

Nanomaterials have nanoscale features that can alter surface area and behaviour. Carbon nanotubes, fullerenes and graphene are examples. Metal–organic frameworks are porous advanced materials. Biomaterials can be metallic, ceramic, polymeric or composite.

**Answer framework:** Connect structure to application carefully; high surface area explains some uses but is not a universal answer. Label material class in comparisons.

**Important point:** Nano describes scale, not a guarantee that a material is safe or suitable for every use.

**Explain Biofuel and green chemistry and state one correct method, application or precaution.−**

Biofuels are fuels derived from biological sources. Green chemistry seeks prevention of waste and safer, resource-conscious chemical design. The syllabus asks for principles, importance and applications.

**Answer framework:** For a process, identify feedstock, energy demand, waste, hazards and potential improvement. Use green-chemistry principles as a checklist.

**Important point:** Sustainable lab practice includes segregation and hazard control; using less of a hazardous reagent is not automatically sufficient.

**Explain Materials laboratory practice and state one correct method, application or precaution.−**

Official experiments include estimation of copper/zinc in brass, iron in ore and preparation of urea-formaldehyde and phenol-formaldehyde resins. A record connects chemical principle, observations and calculated result.

**Answer framework:** Plan sequence, label samples, keep raw-data table and discuss uncertainty before finalising result.

**Important point:** Resin preparation and metal analysis require PPE, ventilation and supervised waste disposal.

PRACTICE ONLY

# Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

## Part A — short response

1. State one key definition from Module 1: Fundamentals, Solutions and Thermodynamics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Analytical Chemistry, Water and Acids/Bases.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Electrochemistry, Corrosion and Batteries.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Engineering Materials and Green Chemistry.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

## Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

## Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Atoms and molecules; solution concentration; systems, processes and thermodynamic functions; preparation of standard solutions..
2. Develop a complete answer or practical plan for: Volumetric analysis; hardness and water treatment; potable water; pH and pOH; standardisation and estimation..
3. Develop a complete answer or practical plan for: Redox reactions; conductors; electrolysis; Faraday laws; electrochemical cells; corrosion prevention; emerging batteries..

4. Develop a complete answer or practical plan for: Alloys; polymers and rubbers; nanomaterials; MOFs; biomaterials; biofuel; green chemistry..

LAST-MILE REVISION

## Quick Revision

### Module 1: Fundamentals, Solutions and Thermodynamics

Atoms and molecules; solution concentration; systems, processes and thermodynamic functions; preparation of standard solutions.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Analytical Chemistry, Water and Acids/Bases

Volumetric analysis; hardness and water treatment; potable water; pH and pOH; standardisation and estimation.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Electrochemistry, Corrosion and Batteries

Redox reactions; conductors; electrolysis; Faraday laws; electrochemical cells; corrosion prevention; emerging batteries.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Engineering Materials and Green Chemistry

Alloys; polymers and rubbers; nanomaterials; MOFs; biomaterials; biofuel; green chemistry.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

## Glossary

**Important terms from the syllabus**

| Term / topic | One-line meaning |
|---|---|
| Atoms, particles and atomic identifiers | An atom is the smallest chemical unit retaining the identity of an element; a molecule is an associated group of atoms. |
| Solutions, molarity, normality and ppm | A solution contains solute dispersed in solvent. |
| Thermodynamic systems and processes | A system is the selected part for study; surroundings are everything else relevant to it. |
| Solution-preparation and thermal-reaction practice | The official work includes 0. |
| Volumetric analysis and indicators | Titration finds concentration by reacting measured volumes to an endpoint identified using a suitable indicator. |
| Hardness, potable water and treatment | Hard water contains dissolved salts that interfere with soap action; temporary and permanent hardness have different origin/treatment. |
| Arrhenius acids, bases, pH and pOH | Arrhenius acids produce hydrogen ions in aqueous solution and bases produce hydroxide ions. |
| Quantitative and qualitative analysis practice | Official activities include HCl and EDTA standardisation, estimation of NaOH/KOH, total hardness by EDTA and pH determination by meter, universal indicator and paper. |
| Redox, conductors and electrolysis | Oxidation and reduction occur together in a redox reaction. |
| Electrochemical cells and applications | A Daniel cell illustrates conversion of chemical energy to electrical energy and uses a salt bridge to maintain charge balance. |
| Corrosion and prevention | Corrosion is deterioration of metal through chemical or electrochemical interaction with environment. |
| Electrochemistry laboratory practice | The official work includes salt bridge and Daniel-cell construction, nickel coating of steel, conductivity observation and electrolysis of water. |
| Alloys, polymers and rubber | Alloying changes useful engineering properties. |
| Nanomaterials, MOFs and biomaterials | Nanomaterials have nanoscale features that can alter surface area and behaviour. |
| Biofuel and green chemistry | Biofuels are fuels derived from biological sources. |
| Materials laboratory practice | Official experiments include estimation of copper/zinc in brass, iron in ore and preparation of urea-formaldehyde and phenol-formaldehyde resins. |

LEARNING RESOURCES

# References

### Officially listed print resources

1. Puri, Sharma and Pathania, Principles of Physical Chemistry.
2. Arthur Israel Vogel, Vogel’s Textbook of Quantitative Chemical Analysis.
3. A. O. Thomas, Practical Chemistry.

### Officially listed web resources

- https://nptel.ac.in/courses/104106119

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